

Learn, Imagine, and Paint: Aesthetic Cognition as Guidance for Robotic Painting

Author Names Omitted for Anonymous Review. Paper-ID [4]

Abstract—Robotic painting requires more than reproducing a target image; it requires grounding visual intentions in physical actions and aesthetic judgment. This paper presents an initial pipeline toward aesthetic cognition-guided robotic painting. The system converts a generated or user-provided image into a paint plan, organizes the plan into executable brush and color batches, and rolls out the resulting trajectories on a LeRobot SO-101 arm. Preliminary experiments demonstrate the feasibility of transferring generated visual content into physical robot motion. While the current system is open-loop, it establishes a physical grounding layer for future work in which robots can reason about aesthetic qualities such as softness, roughness, energy, delicacy, and balance, and use such judgments to guide tool choice, stroke planning, and closed-loop refinement. A demo video is available online.

I. INTRODUCTION

Human painting is not merely the reproduction of an image. Artists continuously reason about visual composition, material behavior, tool choice, and aesthetic qualities such as softness, roughness, energy, delicacy, and balance. A single visual intention can be physically realized in many different ways depending on the tool, stroke motion, contact pressure, and medium. For example, a thin brush may create delicate contours, while a sponge or scraper-like tool may produce more diffuse or textured marks. This coupling between aesthetic judgment and physical tool use makes painting a compelling domain for embodied robotic intelligence.

Recent robotic painting systems have made significant progress in translating high-level user inputs into physical paintings. FRIDA introduced a collaborative robot painting framework that uses language descriptions or images as inputs and optimizes painting plans for physical execution [7]. CoFRIDA extended this direction toward human-robot co-painting by enabling the robot to modify or continue partial human artworks [8]. Spline-FRIDA further studied human-like robot brush stroke trajectories by modeling more diverse spline-based strokes [2]. Learning-based robotic painting has also explored demonstration-based acquisition of continuous stroke-level painting skills [6]. These works demonstrate the promise of integrating generative models, painting representations, and robotic execution. However, many existing pipelines still emphasize target-image reproduction or visual similarity as the primary optimization objective. The robot is usually asked to generate or follow strokes that visually match a desired image, while the higher-level question of how aesthetic judgment can guide physical tool use remains underexplored.

In this work, we use *aesthetic cognition* as a guiding concept for robotic painting. By aesthetic cognition, we refer to the

ability to infer human-perceived aesthetic properties of an artwork and use this understanding to guide physically grounded creative actions. Most existing robot-painting systems evaluate or guide painting through visual similarity to a target image, using image-space feedback, stroke replication, or region-level reconstruction as the main optimization signal [10, 9, 4, 3, 5]. This paradigm is useful for image reproduction, but it does not directly capture the expressive qualities that human viewers often use to interpret artworks. Our work instead aims to give the robot a perceptual evaluation capability: whether a region appears soft or rough, calm or energetic, delicate or visually balanced, and how different tools, pressures, speeds, or trajectories may alter these perceived qualities. However, before such aesthetic reasoning can be used to guide robot behavior, a more basic grounding problem must be addressed: generated visual intentions must first be converted into physically executable robot actions.

At this stage, we present an initial physical grounding pipeline for robotic painting with preliminary experiments to demonstrate the feasibility of robot painting. We implement a lightweight, low-cost, open-loop physical grounding pipeline on a LeRobot SO-101 arm [1] that converts generated or user-provided images into structured paint plans, executable brush/color batches, and robot trajectories, serving as a testbed for future aesthetic-cognition modules.

Our contributions are:

- We present an initial generation-to-execution pipeline that converts generated or user-provided images into structured paint plans and robot-executable trajectories.
- We demonstrate preliminary physical rollout of generated drawing trajectories on a low-cost LeRobot SO-101 arm.
- We position this pipeline as a physical grounding layer for future aesthetic cognition in robotic painting, where human-perceived aesthetic properties can guide tool choice, stroke planning, and closed-loop refinement.

II. METHOD AND PRELIMINARY RESULTS

We implement an initial open-loop robotic painting pipeline that converts generated or user-provided images into executable motions for a LeRobot SO-101 arm. As shown in Fig. 1, the system first normalizes the input image to the physical drawing workspace and extracts visually meaningful contours or stroke-like paths. These paths are represented as a structured paint plan containing geometry, color, and tool information.

The paint plan is then converted into execution batches. Infeasible paths are removed, and the remaining operations

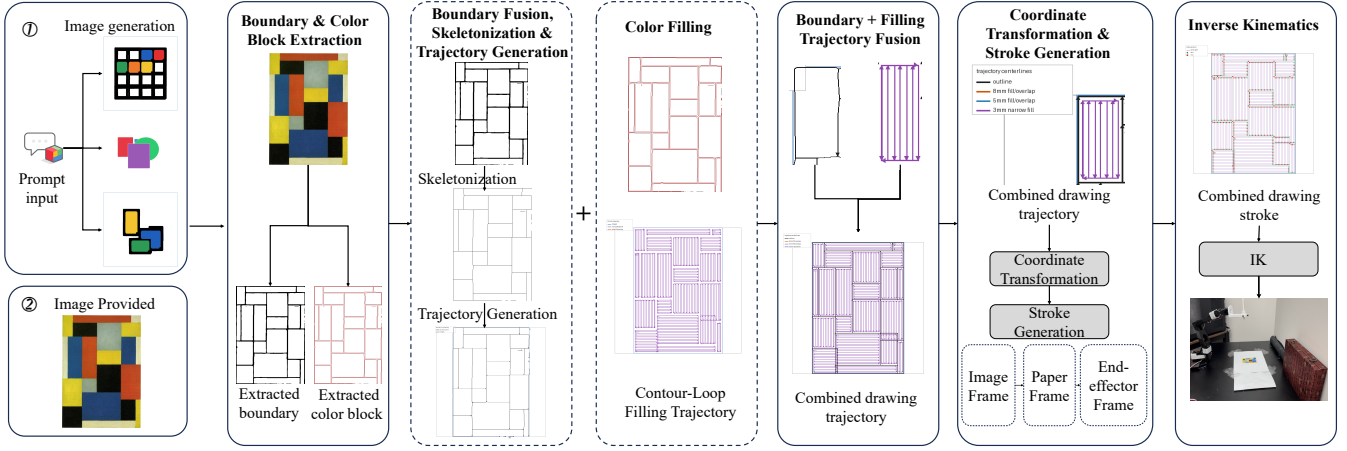


Fig. 1. **Image-to-execution pipeline for robotic painting.** The system takes either a prompt-generated image or a user-provided image as input, extracts color-block boundaries and filled regions, and converts them into robot-executable trajectories. Boundary paths are obtained by binarizing and skeletonizing the extracted boundaries, then tracing connected skeleton pixels under 8-neighborhood to form ordered waypoint sequences, while filling paths are generated by contour-loop filling with longer segments drawn first until the filled area exceeds 90%. The boundary and filling trajectories are fused, transformed from the image frame to the paper and end-effector frames, and converted into stroke commands and joint trajectories through inverse kinematics.

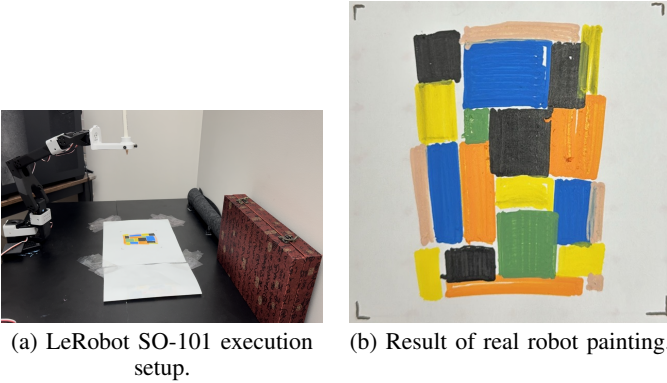


Fig. 2. **Physical execution of generated visual content.** The robot executes the extracted drawing trajectories on a real drawing plane, demonstrating the feasibility of transforming generated images into physically realizable robot motions.

are organized into stroke-level actions, including approach, drawing, and lifting motions. Operations are grouped by brush and color requirements to reduce unnecessary manual changes. During execution, the system maps paper-space coordinates to robot joint targets and streams the resulting commands to the robot to draw on an A4 paper surface.

Preliminary experiments show that simple prompt-generated or reference-based images, such as abstract shapes, simple objects, and stylized drawings, can be converted into robot-executable trajectories and physically rolled out on the LeRobot SO-101 arm, as shown in Fig. 2. Although the current results are proof-of-concept and depend on calibration, tool contact, trajectory smoothness, and surface alignment, they demonstrate a basic physical grounding layer between generated visual plans and embodied robotic painting. Future work will incorporate visual feedback, automatic tool changing, and aesthetic cognition modules for closed-loop refinement.

III. DISCUSSION AND FUTURE WORK

Toward aesthetic cognition for multi-tool robotic painting, we aim to model the perceived aesthetic state of the evolving

artwork rather than treating a generated image only as a fixed reproduction target. This state can then guide future physical mark-making decisions. We represent this state as a compact aesthetic descriptor:

$$z_t^{\text{aesthetic}} = [s_t^{\text{softness}}, s_t^{\text{roughness}}, s_t^{\text{energy}}, s_t^{\text{delicacy}}, s_t^{\text{balance}}].$$

Each component denotes a human-perceived visual quality estimated from the current canvas image I_t . The future control problem can then be formulated as:

$$(I_t, g, \mathcal{T}, z_t^{\text{aesthetic}}) \rightarrow (\text{tool}, a_t),$$

where g is the target aesthetic intention, $\mathcal{T}$ is the available tool set, and a_t is a parameterized painting action. This formulation is intended as the next-stage control objective rather than a module implemented in the current open-loop prototype. Table I illustrates how these aesthetic descriptors can be grounded in perceptual cues and future robot actions.

In summary, this work provides an image-to-trajectory-to-execution testbed for studying how human aesthetic judgment can guide robotic tool selection, stroke planning, and closed-loop refinement.

TABLE I
GROUNDING AESTHETIC DESCRIPTORS IN FUTURE ROBOT ACTIONS.

Aesthetic goal	Perceptual cue	Action/tool implication
softness	diffuse edges, low contrast	sponge-like dabbing; lower pressure
roughness	irregular edges, high texture variation	scraper-like motion; higher pressure
delicacy	thin lines, low overlap, stable contours	thin brush; slower motion
energy	long strokes, high curvature, strong contrast	faster strokes; larger gestures
balance	stable spatial layout, even visual weight	adjust stroke density, color placement, or corrective strokes

REFERENCES

- [1] Remi Cadene, Simon Alibert, Francesco Capuano, Michel Aractingi, Adil Zouitine, Pepijn Kooijmans, Jade Choghari, Martino Russi, Caroline Pascal, Steven Palma, Mustafa Shukor, Jess Moss, Alexander Soare, Dana Aubakirova, Quentin Lhoest, Quentin Gallouédec, and Thomas Wolf. LeRobot: An open-source library for end-to-end robot learning, 2026.
- [2] Lawrence Chen, Peter Schaldenbrand, Tanmay Shankar, Lia Coleman, and Jean Oh. Spline-FRIDA: Towards diverse, humanlike robot painting styles with a sample-efficient, differentiable brush stroke model, 2024.
- [3] Jörg Marvin Gülzow and Oliver Deussen. Region-based approaches in robotic painting. *Arts*, 11(4), 2022. doi: 10.3390/arts11040077.
- [4] Jörg Marvin Gülzow, Liat Grayver, and Oliver Deussen. Self-improving robotic brushstroke replication. *Arts*, 7(4), 2018. doi: 10.3390/arts7040084.
- [5] Songhua Liu, Tianwei Lin, Dongliang He, Fu Li, Ruifeng Deng, Xin Li, Errui Ding, and Hao Wang. Paint transformer: Feed forward neural painting with stroke prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 6598–6607, 2021.
- [6] Younghyo Park, Seunghun Jeon, and Taeyoon Lee. Robot learning to paint from demonstrations. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3053–3060. IEEE, 2022.
- [7] Peter Schaldenbrand, James McCann, and Jean Oh. FRIDA: A collaborative robot painter with a differentiable, real2sim2real planning environment. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2023. doi: 10.1109/ICRA48891.2023.10160702.
- [8] Peter Schaldenbrand, Gaurav Parmar, Jun-Yan Zhu, James McCann, and Jean Oh. CoFRIDA: Self-supervised fine-tuning for human-robot co-painting. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 2296–2302, 2024.
- [9] The e-David Project. The e-David Project: Rediscovering art and artisanship through robotic painting, 2026. URL <https://edavid.de/>. Official project page.
- [10] Patrick Tresset and Frederic Fol Leymarie. Portrait Drawing by Paul the Robot. *Computers & Graphics*, 37(5):348–363, 2013. doi: 10.1016/j.cag.2013.01.012.